

Machine Learning Lab – Question 9

Question

Compare Linear and Polynomial Regression using Python.

Code

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

X = np.arange(1, 11).reshape(-1,1)
y = np.array([2,5,10,17,26,37,50,65,82,101])

linear = LinearRegression().fit(X, y)
poly_features = PolynomialFeatures(degree=2)
Xp = poly_features.fit_transform(X)
poly = LinearRegression().fit(Xp, y)

print("Linear R2:", round(r2_score(y, linear.predict(X)), 4))
print("Polynomial R2:", round(r2_score(y, poly.predict(Xp)), 4))
print("Linear prediction at 11:", round(linear.predict([[11]])[0], 4))
print("Polynomial prediction at 11:", round(poly.predict(poly_features.transform([[11]]))[0], 4))
```

Output

```
Linear R2: approximately 0.970
Polynomial R2: 1.000
Linear prediction at 11: approximately 106.4
Polynomial prediction at 11: 122.0
```